Certified Frobenius Orientation from `proof2` and PARI/GP

An informal corollary note for the `frob-orientation-lean` record

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Bottom line. For every prime q appearing in `PARI_GP_Certificate_Proof.txt` (in particular, for the current certified range produced by the attached script up to $q < 100000$), the original Frobenius Orientation Theorem with the *positive* choice of a follows informally from two ingredients:

1. the kernel-checked Lean deduction in `proof2_frobenius_orientation_aristotle.lean`, and
2. the exact cyclotomic witness data generated by the PARI/GP certificate.

What is not yet internalized in Lean is only the import of the certificate itself, not the mathematical deduction from that certificate to the Frobenius conclusion.

Purpose

This note records a clean way to place the original Frobenius Orientation Theorem on solid footing without waiting for a fully internalized Lean certificate. The strategy is simple:

1. interpret the final Lean file `proof2_frobenius_orientation_aristotle.lean` as a conditional theorem whose only serious hypothesis is a cyclotomic-period witness package;
2. observe that the updated PARI/GP script constructs exactly that witness package by exact arithmetic in the cyclotomic quotient ring;
3. deduce the normalized Frobenius Orientation statement from Lean, and then pass from the normalized sign convention $a \equiv 1 \pmod 3$ to the original statement with the positive square root.

The result is an *informal mathematical corollary* of the Lean theorem plus the PARI/GP certificate. That is enough for the present record and, in particular, for repository notes or a project memorandum.

What `proof2` proves

Let

$$f(x) = x^3 - qx - q, \qquad \chi_q(n) = n^{(q-1)/3} \pmod q.$$

In the Lean file, the two key theorems are `irreducibility_criterion` and, separately, `frobenius_orientation`. Their content is this.

Suppose one is given a witness package $h_{\text{valid}}(q, a)$ consisting of:

- a cyclotomic field $L/\mathbf{Q}$ of level q,
- a primitive root g mod q,
- the normalization $a \equiv 1 \pmod 3$,
- the period equation for the Gaussian-period root α,
- the defining relation for the companion root β, and
- the orientation condition $\chi_q(g) = s_2$.

Then Lean proves:

(1) if $\chi_q(N) \neq 1$, then f is irreducible modulo N;

(2) if f is irreducible modulo N, then

$$\chi_q(N) = s_1 \Longrightarrow \alpha^N = \gamma, \qquad \chi_q(N) = s_2 \Longrightarrow \alpha^N = \beta.$$

The proof is genuinely formal Lean mathematics. What remains external is only the production of $h_{\text{valid}}(q, a)$.

What the PARI/GP certificate supplies

The file `PARI_GP_Certificate_Proof.txt` does precisely that witness construction. It runs over primes $q < 100000$ with $4q - 27$ a square, defines

$$a = \pm\sqrt{4q - 27}$$

with the sign chosen so that $a \equiv 1 \pmod 3$, chooses a primitive root g mod q, and, if necessary, replaces g by g^{-1} so that

$$g^{(q-1)/3} = s_2.$$

It then works in the exact cyclotomic quotient ring

$$\mathbf{Q}[x]/(\Phi_q(x))$$

through the PARI object `Mod(x,polcyclo(q))`, so the computation is exact algebra, not floating-point numerics.

With ζ_q represented by the residue class of x, the script defines Gaussian periods

$$\eta_0 = \sum_{j=1}^{(q-1)/3} \zeta_q^{g^{3j}}, \qquad \eta_1 = \sum_{j=1}^{(q-1)/3} \zeta_q^{g^{3j+1}}, \qquad \eta_2 = \sum_{j=1}^{(q-1)/3} \zeta_q^{g^{3j+2}},$$

and then sets

$$\alpha = \eta_0 - \eta_1, \qquad \beta = \eta_1 - \eta_2, \qquad \gamma = \eta_2 - \eta_0.$$

It checks exactly that

$$f(\alpha) = f(\beta) = f(\gamma) = 0, \qquad 2a\beta = 6\alpha^2 - (a+9)\alpha - 4q.$$

Since $\alpha + \beta + \gamma = 0$ is automatic from the definitions, this is exactly the witness package that the Lean theorem asks for.

The output lines in the certificate file have the form

$$q \quad 1 \quad 0 \quad 0 \quad 0 \quad 0,$$

meaning:

- the chosen sign satisfies $a \equiv 1 \pmod 3$,
- the three period differences α, β, γ satisfy $f = 0$, and
- the defining relation for β also vanishes.

Normalized corollary from Lean plus PARI/GP

We now record the statement that follows directly from the two files.

Corollary 1 (Normalized Frobenius Orientation Theorem in the certified range)**.** *Let q be a prime appearing in* `PARI_GP_Certificate_Proof.txt`*, and let $a \in \mathbf{Z}$ be the sign chosen by the certificate, so that*

$$4q = a^2 + 27, \qquad a \equiv 1 \pmod 3.$$

Let N be an odd prime with $N \neq q$ and $\gcd(a, N) = 1$. Put

$$s_1 \equiv \frac{-a-3}{6} \pmod q, \qquad s_2 \equiv \frac{a-3}{6} \pmod q,$$
$$c \equiv N^{(q-1)/3} \pmod q, \qquad f(x) = x^3 - qx - q.$$

Then:

(1) If $c \not\equiv 1 \pmod q$, then f is irreducible modulo N.

(2) Suppose f is irreducible modulo N. Let K be a splitting field of f over $\mathbf{F}_N$. Let $\alpha \in K$ be a root of f, and define $\beta, \gamma \in K$ by

$$2a\beta = 6\alpha^2 - (a+9)\alpha - 4q, \qquad \alpha + \beta + \gamma = 0.$$

Then

$$\alpha^N = \gamma \quad \text{if } c \equiv s_1 \pmod q, \qquad \alpha^N = \beta \quad \text{if } c \equiv s_2 \pmod q.$$

Proof. For each such q, the PARI/GP file constructs the witness package required by the Lean theorem: a cyclotomic field, an oriented primitive root g, the period equation, the relation defining β, and the normalization $\chi_q(g) = s_2$. In the notation of the Lean file, $\chi_q(N) = N^{(q-1)/3} \bmod q = c$. Therefore the Lean theorem `irreducibility_criterion` gives part (1), and the Lean theorem `frobenius_orientation` gives part (2). The deduction from the certificate to the stated conclusion is informal here, but mathematically straightforward. □

Passage from the normalized sign to the original positive sign

The original formulation uses the positive integer $A > 0$ satisfying

$$4q = A^2 + 27.$$

Because $3 \nmid A$, one has either $A \equiv 1 \pmod 3$ or $A \equiv 2 \pmod 3$. If $A \equiv 1 \pmod 3$, then the normalized theorem already uses $a = A$. The only interesting case is $A \equiv 2 \pmod 3$, where the certificate uses $a = -A$.

Lemma 2 (Sign-change identities)**.** *Let $A > 0$ satisfy $4q = A^2 + 27$, and suppose the normalized sign is $a = -A$. Then*

$$s_1(-A) = s_2(A), \qquad s_2(-A) = s_1(A).$$

If β_A, γ_A are defined from a root α by

$$2A\beta_A = 6\alpha^2 - (A+9)\alpha - 4q, \qquad \alpha + \beta_A + \gamma_A = 0,$$

and β_{-A}, γ_{-A} are defined similarly with $-A$ in place of A, then

$$\beta_{-A} = \gamma_A, \qquad \gamma_{-A} = \beta_A.$$

Proof. The identities for s_1 and s_2 are immediate from the definitions. For the second part, compute

$$2(-A)\gamma_A = -2A(-\alpha - \beta_A) = 2A\alpha + 2A\beta_A = 6\alpha^2 - ((-A)+9)\alpha - 4q.$$

So γ_A satisfies the defining equation for β_{-A}, hence $\beta_{-A} = \gamma_A$. Then

$$\gamma_{-A} = -\alpha - \beta_{-A} = -\alpha - \gamma_A = \beta_A.$$

□

Theorem 3 (Original Frobenius Orientation Theorem in the certified range)**.** *Let q be a prime appearing in* `PARI_GP_Certificate_Proof.txt`*, and let $A > 0$ be the positive integer such that*

$$4q = A^2 + 27.$$

Let N be an odd prime with $N \neq q$ and $\gcd(A, N) = 1$. Define

$$s_1 \equiv \frac{-A-3}{6} \pmod q, \qquad s_2 \equiv \frac{A-3}{6} \pmod q,$$
$$f(x) = x^3 - qx - q, \qquad c \equiv N^{(q-1)/3} \pmod q.$$

Then:

(1) If $c \not\equiv 1 \pmod q$, then f is irreducible modulo N.

(2) Suppose f is irreducible modulo N. Let K be a splitting field of f over $\mathbf{F}_N$. Let $\alpha \in K$ be a root of f, and define $\beta, \gamma \in K$ by

$$2A\beta = 6\alpha^2 - (A+9)\alpha - 4q, \qquad \alpha + \beta + \gamma = 0.$$

Then

$$\alpha^N = \gamma \quad \text{if } c \equiv s_1 \pmod q, \qquad \alpha^N = \beta \quad \text{if } c \equiv s_2 \pmod q.$$

Proof. If $A \equiv 1 \pmod 3$, this is exactly the normalized corollary above. If $A \equiv 2 \pmod 3$, apply the normalized corollary with $a = -A$. By the sign-change lemma,

$$s_1(-A) = s_2(A), \qquad s_2(-A) = s_1(A), \qquad \beta_{-A} = \gamma_A, \qquad \gamma_{-A} = \beta_A.$$

Thus the normalized conclusion

$$\alpha^N = \gamma_{-A} \text{ when } c = s_1(-A), \qquad \alpha^N = \beta_{-A} \text{ when } c = s_2(-A)$$

translates into

$$\alpha^N = \beta_A \text{ when } c = s_2(A), \qquad \alpha^N = \gamma_A \text{ when } c = s_1(A),$$

which is exactly the original positive-sign statement. □

Concluding status statement

The theorem above is not yet a single self-contained Lean theorem, because the PARI/GP certificate has not been imported into Lean as a checked data object. But for every q in the certificate file, there is no remaining mathematical mystery in the deduction:

- the Lean development proves the transfer from an oriented cyclotomic witness to the Frobenius statement, and
- the PARI/GP file supplies that witness by exact cyclotomic calculation.

Accordingly, the original Frobenius Orientation Theorem is on solid footing in the certified range as an informal corollary of a kernel-checked formal argument plus an exact external certificate.